Table 1: Deployment scorecard, realistic deployment (C3-realistic: partial model + 30% parameter error + AR(1) Student-$t$ sensor noise), dev systems, mean over seeds. Columns as in hyperkkl/tree.py Groups: KKL (latent, lora, hyper), DATA-DRIVEN same-data baselines (nlox, koopman_kf, greybox_*), MODEL-BASED filters (the rest). (sequence OOD; div. = max divergence rate over all blocks; AUROC = output-residual alarm). FAILED: sequence-OOD NRMSE above climatology (value next to the system name).

| system | method | id | shape | seq. | unmeas. | p99 | div. | AUROC | $\mu$s/step | status |
|---|---|---|---|---|---|---|---|---|---|---|
| battery (0.765) | latent | 0.1361 | 0.1325 | 0.1350 | 0.1432 | 0.2184 | 0.00 | 0.955 | 1,606 | |
| | HyperKKL | 0.0506 | 0.0568 | 0.0642 | 0.0698 | 0.1495 | 0.00 | 0.940 | 1,665 | |
| | dense (abl.) | 0.0478 | 0.0510 | 0.0618 | 0.0665 | 0.0954 | 0.00 | 0.945 | 1,171 | |
| | NLOX | 0.1589 | 0.1902 | 0.2837 | 0.3521 | 0.3630 | 0.00 | 0.780 | 1,041 | |
| | EKF | 0.3725 | 0.3304 | 0.2959 | 0.3365 | 0.4376 | 0.00 | 0.878 | 11,405 | |
| | UKF | 0.3709 | 0.2985 | 0.2925 | 0.3321 | 0.4353 | 0.00 | 0.824 | 6,565 | |
| | PF | 0.3765 | 0.3042 | 0.2940 | 0.3288 | 0.4374 | 0.00 | 0.943 | 5,898 | |
| | KKL-MB | 1.3581 | 1.2096 | 1.4683 | 1.5779 | 2.1881 | 0.00 | 0.842 | 16,875 | FAILED |
| | EKF-Hu | 0.3730 | 0.3211 | 0.2951 | 0.3203 | 0.4358 | 0.00 | 0.882 | 11,785 | |
| | UKF-Hu | 0.3720 | 0.2997 | 0.2942 | 0.3201 | 0.4352 | 0.00 | 0.871 | 7,423 | |
| | EKF-or | 0.3715 | 0.3235 | 0.2929 | 0.3310 | 0.4347 | 0.00 | 0.862 | 12,494 | |
| | UKF-or | 0.3702 | 0.2980 | 0.2901 | 0.3275 | 0.4329 | 0.00 | 0.819 | 5,066 | |
| | Koopman KF | 0.1022 | 0.1078 | 0.0961 | 0.0815 | 0.3155 | 0.00 | 0.736 | 23,552 | |
| | GB-EKF | 0.0185 | 0.0206 | 0.0217 | 0.0224 | 0.0392 | 0.00 | 0.928 | 22,250 | |
| | GB-UKF | 0.0191 | 0.0216 | 0.0222 | 0.0233 | 0.0401 | 0.00 | 0.930 | 8,056 | |
| | GB-EKF-or | 0.0185 | 0.0206 | 0.0214 | 0.0221 | 0.0390 | 0.00 | 0.925 | 21,430 | |
| bioreactor (0.708) | latent | 0.0614 | 0.0561 | 0.0702 | 0.0859 | 0.1097 | 0.00 | 0.889 | 1,855 | |
| | HyperKKL | 0.0313 | 0.0350 | 0.0987 | 0.1023 | 0.2843 | 0.00 | 0.890 | 1,612 | |
| | dense (abl.) | 0.0328 | 0.0313 | 0.0922 | 0.0991 | 0.2431 | 0.00 | 0.911 | 1,163 | |
| | NLOX | 0.0171 | 0.0211 | 0.0389 | 0.0334 | 0.0484 | 0.00 | 0.885 | 1,000 | |
| | EKF | 5.7330 | 5.7218 | 5.7313 | 8.0282 | 5.7950 | 0.00 | – | 8,679 | FAILED |
| | UKF | 5.7103 | 5.6993 | 5.7076 | 7.9949 | 5.7692 | 0.00 | – | 3,341 | FAILED |
| | PF | 5.7978 | 5.7874 | 5.8001 | 8.1239 | 5.8636 | 0.00 | – | 3,120 | FAILED |
| | KKL-MB | 1.8498 | 1.9388 | 2.9636 | 3.8966 | 3.7956 | 0.00 | 0.863 | 11,992 | FAILED |
| | EKF-Hu | 5.7229 | 5.7118 | 5.7243 | 8.0155 | 5.7882 | 0.00 | – | 6,608 | FAILED |
| | UKF-Hu | 5.7220 | 5.7110 | 5.7234 | 8.0143 | 5.7874 | 0.00 | – | 3,292 | FAILED |
| | EKF-or | 2.5540 | 2.5247 | 2.4441 | 3.4564 | 2.6467 | 0.00 | 0.994 | 6,765 | FAILED |
| | UKF-or | 3.0024 | 2.9752 | 2.8975 | 4.0976 | 3.1013 | 0.00 | 0.995 | 2,295 | FAILED |
| | Koopman KF | 0.0402 | 0.0399 | 0.0933 | 0.1287 | 0.1274 | 0.00 | 0.967 | 3,216 | |
| | GB-EKF | 0.0640 | 0.0615 | 0.0649 | 0.0889 | 0.0772 | 0.00 | 0.994 | 11,168 | |
| | GB-UKF | 0.0630 | 0.0607 | 0.0642 | 0.0878 | 0.0760 | 0.00 | 0.995 | 4,632 | |
| | GB-EKF-or | 0.0529 | 0.0528 | 0.0521 | 0.0677 | 0.0618 | 0.00 | 0.986 | 12,120 | |
| duffing (1.71) | latent | 0.0554 | 0.0569 | 0.0824 | 0.1117 | 0.1580 | 0.00 | 0.699 | 1,674 | |
| | HyperKKL | 0.0697 | 0.0760 | 0.0997 | 0.1380 | 0.1196 | 0.00 | 0.686 | 1,314 | |
| | dense (abl.) | 0.0707 | 0.0768 | 0.1072 | 0.1478 | 0.1264 | 0.00 | 0.699 | 1,122 | |
| | NLOX | 0.0901 | 0.0881 | 0.1104 | 0.1521 | 0.1529 | 0.00 | 0.773 | 873 | |
| | EKF | 0.3347 | 0.1204 | 0.2441 | 0.3319 | 0.4500 | 0.00 | 0.752 | 5,369 | |
| | UKF | 0.3343 | 0.1203 | 0.2438 | 0.3318 | 0.4494 | 0.00 | 0.753 | 1,656 | |
| | PF | 0.3988 | 0.1213 | 0.2438 | 0.3322 | 0.4572 | 0.00 | 0.776 | 1,511 | |
| | KKL-MB | 0.1973 | 0.1858 | 0.1957 | 0.2753 | 0.2377 | 0.00 | 0.703 | 11,138 | |
| | EKF-Hu | 0.4071 | 0.1224 | 0.2763 | 0.3485 | 0.5616 | 0.00 | 0.826 | 7,214 | |
| | UKF-Hu | 0.4048 | 0.1221 | 0.2754 | 0.3476 | 0.5595 | 0.00 | 0.825 | 2,698 | |
| | EKF-or | 0.1430 | 0.0958 | 0.1266 | 0.1773 | 0.1948 | 0.00 | 0.673 | 3,685 | |
| | UKF-or | 0.1430 | 0.0958 | 0.1266 | 0.1773 | 0.1948 | 0.00 | 0.673 | 2,011 | |
| | Koopman KF | 0.1595 | 0.1584 | 0.1649 | 0.2316 | 0.1967 | 0.00 | 0.678 | 2,939 | |
| | GB-EKF | 0.0127 | 0.0127 | 0.0144 | 0.0183 | 0.0194 | 0.00 | 0.684 | 7,783 | |
| | GB-UKF | 0.0128 | 0.0128 | 0.0145 | 0.0184 | 0.0195 | 0.00 | 0.684 | 2,984 | |
| | GB-EKF-or | 0.0128 | 0.0128 | 0.0144 | 0.0175 | 0.0194 | 0.00 | 0.679 | 7,880 | |
| lorenz (0.614) | latent | 0.0191 | 0.0176 | 0.0175 | 0.0190 | 0.0309 | 0.00 | 0.941 | 2,088 | |
| | HyperKKL | 0.0110 | 0.0101 | 0.0104 | 0.0109 | 0.0156 | 0.00 | 0.926 | 1,469 | |
| | dense (abl.) | 0.0165 | 0.0159 | 0.0150 | 0.0160 | 0.0223 | 0.00 | 0.930 | 1,420 | |
| | NLOX | 0.0134 | 0.0125 | 0.0129 | 0.0131 | 0.0182 | 0.00 | 0.947 | 1,057 | |
| | EKF | 3.1657 | 3.1673 | 3.1705 | 2.7913 | 3.1959 | 0.00 | – | 7,013 | FAILED |
| | UKF | 3.5217 | 3.5237 | 3.5275 | 3.1037 | 3.5576 | 0.00 | – | 1,840 | FAILED |
| | PF | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 1.00 | – | 1,923 | FAILED |
| | KKL-MB | 0.0273 | 0.0254 | 0.0257 | 0.0281 | 0.0397 | 0.00 | 0.937 | 12,153 | |
| | EKF-Hu | 3.2244 | 3.2236 | 3.2337 | 2.8441 | 3.4158 | 0.03 | 0.996 | 6,367 | FAILED |
| | UKF-Hu | 3.5837 | 3.5828 | 3.5947 | 3.1600 | 3.8092 | 0.03 | 0.996 | 2,714 | FAILED |
| | EKF-or | 0.6663 | 0.6691 | 0.6722 | 0.6107 | 0.7029 | 0.00 | – | 5,113 | FAILED |
| | UKF-or | 0.6478 | 0.6505 | 0.6535 | 0.5945 | 0.6839 | 0.00 | – | 1,919 | FAILED |
| | Koopman KF | 0.0252 | 0.0236 | 0.0242 | 0.0206 | 0.0366 | 0.00 | 0.921 | 4,239 | |
| | GB-EKF | 0.1112 | 0.0763 | 0.1067 | 0.0512 | 1.1442 | 0.00 | 0.995 | 11,842 | |
| | GB-UKF | 0.0863 | 0.0713 | 0.1017 | 0.0478 | 0.8615 | 0.00 | 0.995 | 4,766 | |
| | GB-EKF-or | 0.2184 | 0.2156 | 0.2160 | 0.2217 | 0.2254 | 0.00 | – | 12,617 | |
| quad_tank (0.581) | latent | 0.3311 | 0.3315 | 0.3421 | 0.4799 | 0.4528 | 0.00 | 0.960 | 2,119 | |
| | HyperKKL | 0.0245 | 0.0257 | 0.0441 | 0.0538 | 0.0628 | 0.00 | 0.963 | 1,559 | |
| | dense (abl.) | 0.0229 | 0.0244 | 0.0436 | 0.0536 | 0.0625 | 0.00 | 0.965 | 1,455 | |
| | NLOX | 0.0890 | 0.0968 | 0.1540 | 0.1835 | 0.1938 | 0.00 | 0.940 | 746 | |
| | EKF | 4.7397 | 4.6491 | 4.6806 | 5.4427 | 4.8801 | 0.00 | – | 11,392 | FAILED |

Table 1: (continued)

| system | method | id | shape | seq. | unmeas. | p99 | div. | AUROC | $\mu$s/step | status |
|---|---|---|---|---|---|---|---|---|---|---|
| | UKF | 4.7362 | 4.6462 | 4.6781 | 5.4403 | 4.8771 | 0.00 | – | 4,425 | FAILED |
| | PF | 5.2010 | 5.1278 | 5.1540 | 5.9683 | 5.4029 | 0.00 | – | 4,546 | FAILED |
| | KKL-MB | 0.9938 | 0.8785 | 1.0089 | 1.4042 | 1.3148 | 0.00 | 0.991 | 18,254 | FAILED |
| | EKF-Hu | 4.8023 | 4.7098 | 4.7429 | 5.5171 | 4.9454 | 0.00 | – | 16,030 | FAILED |
| | UKF-Hu | 4.7984 | 4.7066 | 4.7402 | 5.5145 | 4.9423 | 0.00 | – | 5,681 | FAILED |
| | EKF-or | 3.1109 | 3.0471 | 3.0997 | 3.9109 | 3.2578 | 0.00 | – | 11,232 | FAILED |
| | UKF-or | 3.1093 | 3.0456 | 3.0983 | 3.9090 | 3.2562 | 0.00 | – | 5,086 | FAILED |
| | Koopman KF | 0.0428 | 0.0356 | 0.1051 | 0.1303 | 0.1559 | 0.00 | 0.894 | 22,706 | |
| | GB-EKF | 0.0168 | 0.0170 | 0.0170 | 0.0178 | 0.0220 | 0.00 | 0.962 | 16,203 | |
| | GB-UKF | 0.0168 | 0.0170 | 0.0170 | 0.0178 | 0.0220 | 0.00 | 0.962 | 5,174 | |
| | GB-EKF-or | 0.0193 | 0.0196 | 0.0195 | 0.0194 | 0.0250 | 0.00 | 0.952 | 18,338 | |
| vdp (1.16) | latent | 0.0387 | 0.0360 | 0.0408 | 0.0513 | 0.0614 | 0.00 | 0.685 | 1,806 | |
| | HyperKKL | 0.0294 | 0.0265 | 0.0324 | 0.0390 | 0.0469 | 0.00 | 0.700 | 1,519 | |
| | dense (abl.) | 0.0315 | 0.0287 | 0.0334 | 0.0405 | 0.0506 | 0.00 | 0.697 | 1,133 | |
| | NLOX | 0.0412 | 0.0392 | 0.0472 | 0.0586 | 0.0657 | 0.00 | 0.662 | 902 | |
| | EKF | 0.4279 | 0.4479 | 0.4617 | 0.6008 | 0.5003 | 0.00 | 0.861 | 4,741 | |
| | UKF | 0.4279 | 0.4478 | 0.4617 | 0.6008 | 0.5003 | 0.00 | 0.861 | 2,039 | |
| | PF | 0.3613 | 0.3790 | 0.3924 | 0.5514 | 0.4191 | 0.00 | 0.997 | 1,802 | |
| | KKL-MB | 0.1077 | 0.1009 | 0.1247 | 0.1732 | 0.1597 | 0.00 | 0.670 | 15,663 | |
| | EKF-Hu | 0.7247 | 0.7666 | 0.7901 | 0.8661 | 0.9804 | 0.00 | 0.996 | 6,543 | |
| | UKF-Hu | 0.7244 | 0.7663 | 0.7899 | 0.8659 | 0.9801 | 0.00 | 0.996 | 3,356 | |
| | EKF-or | 0.1477 | 0.1483 | 0.1560 | 0.2184 | 0.2109 | 0.00 | 0.626 | 5,345 | |
| | UKF-or | 0.1477 | 0.1483 | 0.1560 | 0.2184 | 0.2109 | 0.00 | 0.626 | 2,172 | |
| | Koopman KF | 0.2150 | 0.2115 | 0.2170 | 0.3052 | 0.3364 | 0.00 | 0.682 | 2,703 | |
| | GB-EKF | 0.0310 | 0.0295 | 0.0388 | 0.0492 | 0.0510 | 0.00 | 0.484 | 6,597 | |
| | GB-UKF | 0.0310 | 0.0296 | 0.0389 | 0.0493 | 0.0511 | 0.00 | 0.484 | 2,722 | |
| | GB-EKF-or | 0.0858 | 0.0833 | 0.0875 | 0.1202 | 0.1401 | 0.00 | 0.691 | 5,349 | |